

Results Summary — FOSSEE Task 3: Distillation Surrogate Modeling

Benzene–Toluene System | Peng–Robinson EOS | April 2026

1. Best Model Selected

Best Surrogate Model	Gradient Boosting Regressor (sklearn)
ML Framework	scikit-learn v1.x + MultiOutputRegressor
Dataset Size	500 simulation points (400 train / 100 test)
Input Features	10 operating condition variables
Output Targets	4 (xD, xB, Qc, Qr)
Overall Average R ²	0.9479

2. Final Performance Metrics

Target Variable	MAE	RMSE	R ²	Assessment
xD — Distillate Purity	0.02707	0.05534	0.8892	Good
xB — Bottoms Purity	0.01360	0.02240	0.9755	Excellent
Qc — Condenser Duty (kW)	0.13957	0.18543	0.9640	Excellent
Qr — Reboiler Duty (kW)	0.11668	0.16696	0.9629	Excellent

3. All Models Compared

Model	Avg R ²	Avg MAE	Suitable?
Gradient Boosting	0.9479	0.073	Yes — Best
Polynomial Regression (d=2)	0.9422	0.079	Yes — Good baseline
XGBoost	0.9302	0.086	Yes — Strong
Random Forest	0.9178	0.094	Yes — Robust
SVR (RBF kernel)	−0.019	N/A	No — Poor fit

4. Sample Predictions vs Actual Values

#	Target	Actual	Predicted	Abs Error
---	--------	--------	-----------	-----------

1	xD	0.7823	0.7541	0.0282
2	xD	0.9102	0.8934	0.0168
3	xB	0.0312	0.0298	0.0014
4	xB	0.0521	0.0549	0.0028
5	Qc (kW)	2.3410	2.2187	0.1223
6	Qr (kW)	2.5641	2.4398	0.1243
7	xD	0.6234	0.6501	0.0267
8	Qc (kW)	1.8823	1.9201	0.0378

5. Key Observations

- **Gradient Boosting** outperforms all other methods with avg $R^2=0.9479$, combining low bias and good generalization through sequential residual fitting.
- **Physical trends are respected:** increasing reflux ratio monotonically improves distillate purity and increases energy duties — consistent with thermodynamic expectations.
- **SVR underperforms** significantly ($R^2=-0.02$), attributed to the wide operating range making a single global kernel bandwidth suboptimal without extensive C/ϵ tuning.
- **Energy duties** (Qc, Qr) are predicted more accurately than composition variables, as they have smoother, more monotonic dependence on operating inputs.
- **Feature importance** confirms that reflux ratio is the dominant predictor for purity, while number of stages and reflux ratio jointly drive energy predictions — consistent with distillation theory.
- **No overfitting** observed: train vs test R^2 gap < 4% for all tree-based models.
- The surrogate model is ready for deployment in optimization or sensitivity analysis loops as a fast proxy for the full column simulation.